Final Project

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For the final project, you are asked to perform a 'mini research' project. Work can be done by yourself or in pairs. You are encouraged to discuss ideas/progress with me while carrying out the project, and must obtain approval for your proposal before continuing. The final project will count for 75% of your final grade (25% comes from the homework assignments).

The 'mini-research' project should be related to one or more of the topics taught in the course. You may suggest a project of your own, or alternatively, discuss a possible direction with me. Note that while the idea presented in your project needs to be non-trivial, I do not expect it to be necessarily novel (although this would be nice). Here are some possible directions you could consider:

- A new setting: Attempting an existing idea in a new, challenging setting. E.g., considering a setting where only limited 3D data is available.
- Suggesting and attempting an algorithmic improvement to an existing idea (E.g., extending a recent paper).
- Combining two ideas together in an interesting way.
- A comparison of a number of existing approaches to a certain problem that goes beyond existing comparisons in a meaningful way.
- Attempting to solve a real world problem using state-of-the-art approaches such as those taught in class.

You are required to provide a short and concise project proposal, which must be approved before you carry out the full project. You are recommended to follow the following template:

- Problem definition: What are you trying to solve?
- Related work: How is it done today? What are the limits of current practice?
- Approach: What is your proposed approach, and why is it better?
- Motivation: Who cares? If it is successful, what difference will it make?
- Validation: What are the "mid-point" (sanity checks or preliminary experiments) and "final" validations to check for success?

Grading

You will be graded based on the following criteria:

- Proposal (10%). Please provide a proposal describing what you intend to do in this work. See the details provided above.
- Introduction and Related Work (10%). Introduce your idea, the motivation, and how it relates to existing works. Make a summary of relevant papers or other relevant approaches that tackle the problem you are considering. Explain how your idea differs from existing ideas and why it is interesting to investigate. You should cite any work you discuss.

- Method (35%). Provide a detailed and formal explanation of your approach. You are encouraged to define your losses, architectures, and empirical/theoretical settings here. Discuss why your approach is sensible compared to possible alternative approaches.
- Evaluation and Results (35%). This section will assess your evaluation of the method. Describe the experiments that you perform and why they are interesting to consider. Note that you *are not* required to run experiments for any baseline methods mentioned or compared to in the paper, but may refer to existing results in the paper for such baselines. For every experiment, provide a clear explanation of your motivation and detailed implementation details, including model architectures, data, pre-processing and augmentation approaches, loss formulations, training methods, and hyperparameter values. Present all your results and discuss them. You may provide both a qualitative and numerical evaluation in the form of tables, graphs, or figures. You can also perform an ablation study to determine the importance of each part of your method or provide visualizations to gain insight into the method you propose.
  A Note on Negative Results: A rigorously evaluated negative result (e.g., proving that a hypothesized algorithmic improvement does not work as intended) is perfectly acceptable and will not penalize your grade. We evaluate your scientific process, methodological rigor, and critical thinking, not simply achieving state-of-the-art metrics.
- Implementation (10%). You may use code available online, but I will only assess the code you write. Hence, you must make a clear distinction between any code taken from online sources and your own. I will assess the correctness, efficiency and generality of your code.

GPU usage. Please use Google Colab or otherwise available GPU resources from the school. If you intend to use your Lab's resources, you must verify this with your advisor.

Submission

The proposal must be submitted and confirmed with me by the end of the semester, 03/07/2026. For the project, the deadline is 06/09/2026 (special exceptions can be made, e.g., you want to extend this to a paper). Please submit your project in a single zip file including:

- A PDF document detailing your final work. You should attempt to divide it into sections such as 'Introduction and Related Work', 'Method', 'Evaluation and Results', and 'Conclusion'. (You may optionally include your original 'Proposal' as an appendix). Note that this is the suggested format, and you are free to use other formats if you feel this is appropriate.
- A code folder containing your implementation, including a `requirements.txt` or `environment.yml` file specifying your dependencies, and a README file that describes exactly how to reproduce the results you discuss in your report.